

Scenario

In this scenario, I have been asked to investigate a recent security incident that compromised some machines. Here, I am responsible for getting the required information from the database for the investigation. I have access to work with the organisation database and the tables it contains.

Here's how I have done this task:

First, I used an **INNER JOIN** to identify which employees are using machines.

Second, I used **LEFT** and **RIGHT** joins to find machines that do not belong to any specific user and users who do not have any specific machine assigned to them.

Finally, I used an **INNER JOIN** to list all login attempts made by all employees.

Step-1

I started with matching employees to their machines. So, the data is in **machines** and **employees** tables. I used SQL inner join to return all records from the **machines** table. Here, both tables have **device_id** which is a common column here. I ran the following query:

```
SELECT *  
  
FROM machines  
  
INNER JOIN employees ON machines.device_id = employees.device_id;
```

```

MariaDB [organization]> SELECT *
-> FROM machines
-> INNER JOIN employees ON machines.device_id = employees.device_id;
+-----+-----+-----+-----+-----+-----+
| device_id | operating_system | email_client | OS_patch_date | emp |
| employee_id | device_id | username | department |
| office |
+-----+-----+-----+-----+-----+-----+
| a320b137c219 | OS 2 | | Email Client 2 | 2021-03-01 |
| 1000 | 1000 | a320b137c219 | elarson | Marketing |
| East-170 |
| b239c825d303 | OS 1 | | Email Client 1 | 2021-03-01 |
| 1001 | 1001 | b239c825d303 | bmoreno | Marketing |
| Central-276 |
| c116d593e558 | OS 3 | | Email Client 1 | 2021-09-01 |
| 1002 | 1002 | c116d593e558 | tshah | Human Resources |
| North-434 |
| d394e816f943 | OS 3 | | Email Client 2 | 2021-03-01 |
| 1003 | 1003 | d394e816f943 | sgilmore | Finance |

```

Step – 2

Next, I returned the information on all machines and the employees who have machines. At this stage, I used **left join** and **right join** on the **employees** and **machines** tables. The result, here, will include all records from one or the other table. I linked these tables using the common **device_id** column. So, my SQL query was:

```
SELECT *
```

```
FROM machines
```

```
LEFT JOIN employees ON machines.device_id = employees.device_id;
```

```

MariaDB [organization]> SELECT *
-> FROM machines
-> LEFT JOIN employees ON machines.device_id = employees.device_id;
+-----+-----+-----+-----+-----+-----+
--+-----+
| device_id | operating_system | email_client | OS_patch_date | emp
loyee_id | employee_id | device_id | username | department
| office |
+-----+-----+-----+-----+-----+-----+
--+-----+
| a320b137c219 | OS 2 | Email Client 2 | 2021-03-01 |
1000 | 1000 | a320b137c219 | elarson | Marketing
| East-170 |
| b239c825d303 | OS 1 | Email Client 1 | 2021-03-01 |
1001 | 1001 | b239c825d303 | bmoreno | Marketing
| Central-276 |
| c116d593e558 | OS 3 | Email Client 1 | 2021-09-01 |
1002 | 1002 | c116d593e558 | tshah | Human Resources
| North-434 |
| d394e816f943 | OS 3 | Email Client 2 | 2021-03-01 |
1003 | 1003 | d394e816f943 | sgilmore | Finance
| South-153 |
| e218f877g788 | OS 2 | Email Client 1 | 2021-09-01 |
1004 | 1004 | e218f877g788 | eraab | Human Resources
| South-127 |

```

Here, all records from the table referenced after **FROM** and before **LEFT JOIN** are included in the result. In this case, all records from the **machines** table are included, regardless of whether they are assigned to an employee or not.

Next, I connected the **machines** and **employees** tables through a right join like this:

```
SELECT *
```

```
FROM machines
```

```
RIGHT JOIN employees ON machines.device_id = employees.device_id;
```

```
MariaDB [organization]> SELECT *
-> FROM machines
-> RIGHT JOIN employees ON machines.device_id = employees.device_id;
```

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department	office
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	elarson	Marketing	East-170
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing	Central-276
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources	North-434
d394e816f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e816f943	sgilmore	Finance	South-153

Final - step:

To continue investigating the security incident, I retrieved the information on all employees who have made login attempts. I used **inner join** on the **employees** and **log_in_attempts** tables, linking them on the common **username** column. I ran the following query:

```
SELECT *
```

```
FROM employees
```

```
INNER JOIN log_in_attempts ON employees.username = log_in_attempts.username;
```

```
MariaDB [organization]> SELECT *
-> FROM employees
-> INNER JOIN log_in_attempts ON employees.username = log_in_attempts.username;
```

employee_id	device_id	username	department	office	event_id	username	login_date	login_time	country	ip_address	success
1032	g773h303i639	jrafael	Information Technology	Central-309	1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	0
1026	a998b568c863	apatel	Human Resources	West-320	2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
1031	f419g188h578	dkot	Marketing	West-408	3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	0
1031	f419g188h578	dkot	Marketing	West-408	4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
1032	g773h303i639	jrafael	Information Technology	Central-309	5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0

Conclusion

I retrieved the necessary information above for the security team now. They can use this login information to investigate further.